A Few-Shot SAR Target Recognition Method by Unifying Local Classification With Feature Generation and Calibration

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***Abstract*— Recently, metric-based meta-learning has been widely adopted to solve few-shot synthetic aperture radar (SAR) target classification, where the global features are employed to measure the similarity of each sample and the class prototypes. However, due to the high inter-class similarity of SAR images, the global features may suppress the detailed information in local features beneficial for SAR target classification. Besides, the prototypes obtained from a few SAR support samples at sparse azimuth angles tend to be biased. To tackle the above problems, we first integrate a local feature classification module into meta-learning and propose a multi-scale local classification network (MLC-Net) to enhance the target's critical local detail features. Then, the feature generation and calibration network (FGC-Net) is proposed to generate SAR support features at the full range of azimuth angles to compensate for the real support features extracted by MLC-Net. Specifically, FGC-Net consists of a feature generative adversarial network (FGAN) and an adaptive feature calibration module (AFCM). FGAN is designed to generate support features for each class under the full range of azimuth angles. AFCM is proposed to calibrate the generated support features by adaptively re-weighting the generated and real support features of the same class. Finally, we devise the prototype mean square error (PMSE) loss and the central constraint (CC) loss to further narrow the distribution margin between the generated and real support features. FGC-Net and MLC-Net are unified for end-to-end training, resulting in consistent performance gains in feature generation and few-shot classification. Extensive experiments on the moving and stationary target acquisition and recognition (MSTAR) benchmark dataset under different few-shot settings demonstrate the effectiveness of our proposed method.**

***Index Terms*— Calibration, feature generation, few-shot, local feature classification, synthetic aperture radar (SAR), target classification.**

I. Introduction

SYNTHETIC aperture radar (SAR) is widely used in resource exploration and military surveillance for its day-and-night, all-weather working advantages. SAR target classification is one of the critical technologies for the intelligent interpretation of SAR images. Recently, increasing attention has been paid to SAR target classification methods based on convolutional neural networks (CNNs) [1], [2], [3], [4], [5], which rely on vast amounts of accessible labeled data. However, obtaining key SAR targets and their high-quality SAR images in realistic scenarios, such as noncooperative targets, is difficult. Besides, since SAR images reflect the radar backscattering from the ground objects, the targets in SAR images are visually different from the actual targets, resulting in the difficulty of manually labeling the target classes of SAR images. Therefore, the performance of CNN-based SAR target classification methods [1], [2], [3], [4], [5] is limited by the scarce labeled SAR images. The research on SAR target classification with a few labeled images [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19] is desirable.

Data augmentations [1], [6], [20], [21], [22] (e.g., flips, rotations, translations, etc.) are usually utilized to scale up the training set to address the problem of a few labeled SAR images. However, the limited diversity of the augmented images prevents the incorporation of additional rich target information into the classification network. On the other hand, scholars have designed lightweight networks [1], [23] for SAR data to mitigate the risk of severe overfitting due to a small number of labeled images, but this cannot fundamentally solve the problem. In the past few years, transfer learning-based methods [24], [25], [26], [27], [28], [29], [30] have been proposed to tackle the SAR target classification with a few labeled data, which are dedicated to transferring the rich knowledge learned from complementary data (e.g., SAR data of other sources [24], [29], simulated data [25], [26], [27], [28], [30], etc.) obtained from other source domains to the target domain. However, when the number of labeled samples in the target domain decreases sharply (e.g., only five labeled samples per class), the classification performance of the above methods degrades.

Few-shot learning (FSL) based on meta-learning [31] has recently received growing attention. It aims to learn a classifier from a source domain containing abundant labeled images and generalize it to an unseen target domain with a few labeled images. Existing mainstream meta-learning methods roughly fall into two types: optimization-based [32], [33], [34], [35] and metric-based [36], [37]. Optimization-based meta-learning aims to obtain suitable initialization parameters for the network. On the other hand, the metric-based meta-learning approaches are designed to learn an excellent embedding space

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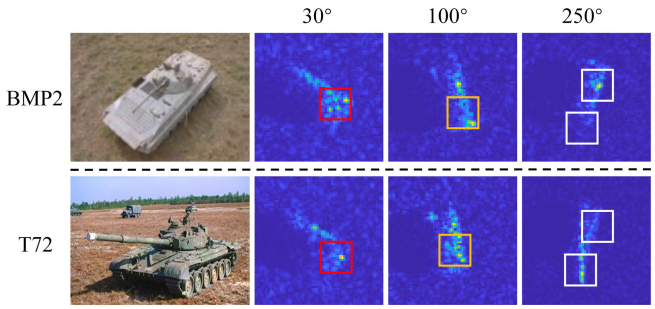


Fig. 1. SAR images of BMP2 and T72 targets under the azimuth angles of 30°, 100°, and 250°, and their respective optical photographs. The key local detail features that distinguish BMP2 from T72 at the same azimuth angles are marked with boxes of the same color.

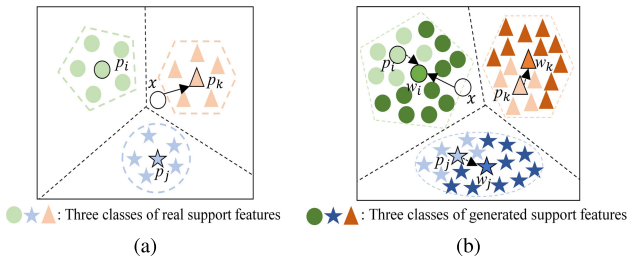


Fig. 2. Schematic illustration of the three-class classification of SAR targets in the embedding space. (a) Classification under the biased prototypes, where p_i, p_j, and p_k represent the biased prototypes of three classes determined by the real support features at some non-continuous azimuth angles, and x denotes the target feature to be classified with real class i. (b) Classification under the modified unbiased prototypes. w_i, w_j, and w_k are unbiased prototypes, which are calculated using the generated support features at the full range of azimuth angles and the real support features together.

and then to predict the class labels of the images by comparing the correlations between the image features and the prototypes of each class. Typically, the prototypes are the average features of the support images (i.e., labeled images) in each class [36]. Since training the optimization-based meta-learning models is time-consuming, metric-based meta-learning methods [6], [7], [8], [9], [10], [11], [12], [13], [14], [15] are broadly utilized in the few-shot SAR target classification field.

Accordingly, this article focuses on metric-based meta-learning to solve the few-shot SAR target classification. Despite the promising improvements in existing few-shot SAR target classification models [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], some issues have not been considered.

1) *Large Inter-Class Similarity:* Because of the unique imaging mechanism, as shown in Fig. 1, only slight local differences exist in SAR images of different classes of targets, which are the key to recognizing the targets. Thus, directly calculating the similarity between the global features would obscure the contribution of the rich local detail features to the classification.
2) *Biased Prototypes:* SAR images at different azimuth angles contain rich information about the target. However, due to the restriction of the observation conditions, we may only get labeled SAR images at some noncontinuous azimuth angles in the target domain, resulting in a bias in the calculated prototypes and degraded performance in few-shot SAR target classification. A schematic illustration of the three-class classification of SAR targets in the embedding space is shown in Fig. 2(a). As shown in Fig. 2(a), the target feature x, which belongs to class i, is misclassified as class k under the biased prototypes.

Nowadays, diverse methods of fine-grained few-shot classification [38], [39], [40] have been proposed to solve the problem of large inter-class similarity of targets. Since the targets have an overall similarity in appearance, these methods mentioned above attend to local details on the images to emphasize the class-wise discriminative features, thus allowing the model to recognize similar targets. For example, Hao et al. [38] refined image-level features into multiple local region features to calculate their similarity, thus enhancing the effectiveness of local detail features in classification. Wang et al. [39], to improve the performance of fine-grained few-shot classification, extracted the part-level features of the targets and enforced the network to learn the differences between the parts of different targets. Accordingly, inspired by fine-grained classification methods, we integrate local feature classification methods into meta-learning and present a high-performance few-shot classifier called multi-scale local classification network (MLC-Net) to solve the problem of large inter-class similarity of SAR targets.

Regarding the second issue, generative adversarial networks (GANs) [41], [42], [43], [44], [45], [46], [47], [48] have been used in recent years to generate diverse SAR images to enrich the representation of SAR targets. Even though the current SAR image generation methods [42], [43], [44], [48] can control the azimuth angles and categories of the generated images, visible disparities persist between the strong scattering centers of the generated images and the corresponding real images, which indicates that the SAR image generation task is challenging. Furthermore, for existing approaches [41], [42], [43], [44], [45], [46], [47], [48], since the training of GAN and the downstream classification are independent stages that are not directly related, the generated SAR images do not provide optimal assistance for the downstream target recognition task. In the optical data domain, feature generation [49], [50], [51], [52], [53] has been researched to overcome the data scarcity in few/zero-shot learning, as feature generation is much easier to implement than high-quality image generation, and features are direct determinants of classification performance. Hence, considering the complexity of the SAR image generation task and the performance improvement of the downstream few-shot classification task, we propose a unified framework in which the feature generation and calibration network (FGC-Net) and the few-shot classifier MLC-Net are trained in an end-to-end manner. In the proposed method, the support features generated by FGC-Net under the full range of azimuth angles compensate for the real support features, thus correcting the prototypes and finally improving the few-shot classification performance, as shown in Fig. 2(b).

The presented FGC-Net contains a feature GAN (FGAN) and an adaptive feature calibration module (AFCM). Specifically, FGAN is first devised to generate support features under the control information of class label and azimuth angle.

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Next, driven by the self-attention mechanism [54], AFCM is proposed to utilize the real support features of the same class to calibrate the generated support features. Finally, with the additional constraints of the designed prototype mean square error (PMSE) loss and the central constraint (CC) loss, FGC-Net can generate support features for each class of SAR targets at the full range of azimuth angles, which are distributed consistently with the real support features. MLC-Net and FGC-Net constitute a novel metric-based meta-learning framework for few-shot SAR target classification, called local classification with feature generation and calibration (LcFGC).

Contributions of this article are summarized as follows.

1) MLC-Net, a few-shot classifier with high performance, is proposed to enhance key local detail features of targets on SAR images.
2) We devise FGC-Net, which is dedicated to generating high-quality support features at the full range of azimuth angles guided by MLC-Net to eliminate the bias of the prototypes.
3) Extensive experiments on the moving and stationary target acquisition and recognition (MSTAR) dataset [55] validate the excellent performance of the proposed LcFGC in the few-shot SAR target classification.

The rest of this article is organized as follows. Section II describes the related work. A detailed description of LcFGC is provided in Section III. The training details and experimental results are given in Section IV. Finally, Section V provides a brief conclusion.

II. Related Work

A. Definition of FSL

Unlike the dataset split in traditional classification, FSL typically consists of a training set from the source domain, and a support set and a test set belonging to the target domain. Notably, the class labels in the source and target domains are disjoint, while the support set and the test set share the same class labels. Given a W-way S-shot few-shot classification setting, there are W classes in the support set with S (usually $S \leq 10$) samples per class.

FSL adopts an episodic training strategy [31] so that prior knowledge learned from the source domain can be generalized to the unseen target domain. Fig. 3 depicts the episodic training process of the three-way one-shot few-shot classification setting. Concretely, the training set is randomly sampled into numerous auxiliary few-shot classification tasks $\{T_i, i = 1, \ldots, n\}$. In each T_i, we randomly select three classes and one sample per class from the training set to form the support set S_i and select a batch of samples from the remaining samples of the selected three classes to form the query set Q_i. Given the selected S_i and Q_i, the query samples are classified according to the support samples, and the classification loss of query samples is responsible for updating the classifier's parameters.

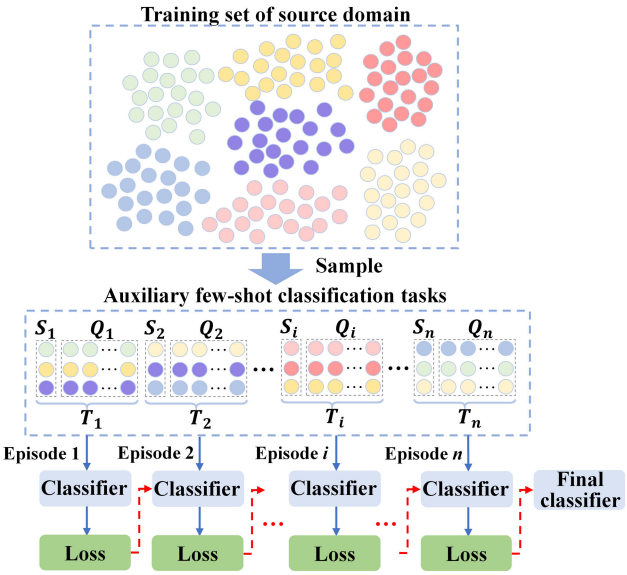


Fig. 3. Episodic training strategy of the few-shot classification in three-way one-shot case. Samples of the same class are marked by circles of the same color. T_i represents the few-shot classification task in the ith episode, and S_i and Q_i denote the support set and query set of T_i, respectively.

B. Metric-Based Meta-Learning for Few-Shot SAR Target Classification

The scarcity of training samples is the main restriction for CNN-based SAR target classification. Due to the advantages of simplicity and efficiency, metric-based meta-learning [6], [7], [8], [9], [10], [11], [12], [13], [14], [15] is widely used in the field of few-shot SAR target classification. The metric-based meta-learning approaches aim to learn an excellent embedding space in which the samples of the same class are mapped into a cluster, and the class labels of samples are predicted by finding the nearest prototypes. For instance, Li et al. [7] integrated the pretraining strategy and improved the structure of the relation network (RN) [37], striving to measure the relationship between sample features and prototypes better. In [15], the triplet loss was applied to constrain sample features of the same class to be close to each other in the embedding space while sample features of different classes are far from each other. Similar to [7] and [15], a weighted distance classifier [12] was presented to adaptively weight the similarity scores calculated by the prototypical network (PN) [36], making the margins larger between the features of different classes. Apart from utilizing the Euclidean distance-based nonparametric classifier, another classifier following the transductive inference [10] was designed to predict the SAR images' class labels jointly. To effectively evaluate the correlation between the sample features to enhance the few-shot SAR target classification, Zhang et al. [11] fully explored the domain knowledge of SAR images and improved the Earth mover's distance.

In the aforementioned metric-based meta-learning approaches, the class labels of SAR images are obtained mainly by computing the similarity between their embedded features and the class prototypes. However, the credibility of the prototypes, calculated from a few support images, is mostly not considered. Ren et al. [16] proposed to update the prototype using the test features jointly with the support features. First, the probability that the test sample belongs to each class is predicted with the trained network. Then,

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test samples whose probability of being predicted as the ith class is greater than 0.5 are selected to calculate the prototype jointly with the support samples of the ith class. However, the network's performance severely affects the class prediction of the test features, thus influencing the reliability of the prototype. Yang et al. [18], assigned each novel class prototype with K most similar prototypes of the base class, for using the base class information to correct the prototypes of the novel class. In this article, FGC-Net is proposed, dedicated to generating the support features under the full range of azimuth angles to calibrate the prototypes together with the real support features.

C. Data Augmentation for FSL

An intuitive method to solve FSL is to augment the training images as much as possible. The existing data augmentation methods are broadly divided into two categories: 1) geometric transformation-based and 2) generative model-based.

1) Geometric Transformation-Based: In early studies [1], [6], [20], [21], [22], geometric transformations were often utilized to augment additional training images, such as rotation, translation, flip, etc. For instance, Chen et al. [1] performed a random translation operation on the training images. Yang et al. [6] used various data augmentation methods with power transformation, noise addition, and flip and rotate operations to scale up the dataset. Apart from translation and adding speckle noise, Ding et al. [20] presented the pose synthesis to generate SAR images under unseen azimuth angles. Although the above methods significantly increase the training images, the diversity of these augmented training images is limited.

2) Generative Model-Based: Generating visually realistic images utilizing generative models, such as variational autoencoders (VAEs) [56], [57] and GANs [58], [59], [60], [61], [62], etc., to assist in training classifiers has become a hot topic. Contrary to other generative models, the adversarial training between the generator and the discriminator allows GANs to learn the distribution of the training data, resulting in more explicit and realistic images. Thus, scholars widely adopt GAN to solve FSL in optical data regimes. For example, Hong et al. [62] proposed F2GAN to generate clear images by fusing the high-level features of conditional images, thus facilitating classification performance in the few-shot case. Gu et al. [60] proposed LoFGAN, a generative model based on a local feature fusion strategy, to ensure the authenticity and reliability of generated images to solve FSL. In [61], the meta image reinforcing network is proposed to fuse the images generated by GAN with the real images, thus producing hybrid images to improve the performance of few-shot classification. With the development of GAN, GAN-based generation methods [41], [42], [43], [44], [45], [46], [47], [48] are proposed to expand the limited SAR training data. For instance, Du et al. [41] generated SAR images by inputting the encoded features into the generator to enhance the quality of generated SAR images. Song et al. [42] introduced the label and orientation angle as conditional information to improve the diversity of the generated SAR images. PeaceGAN [44] was proposed with a pose estimator and an auxiliary classifier to generate SAR images with faithful pose angles of desired classes.

To save computational costs, more and more attention has been paid to using GANs to generate optical image features [49], [50], [51], [52], [53] in recent years. Li et al. [50] argued that features are a direct determinant of classifier performance and proposed a new generative model unifying a classifier, hoping to generate useful features for classification. Due to the stable training of VAE, Xian et al. [51] integrated VAE and GAN to generate image features of each class. To gather reliable prototypes, a cross-model feature generation framework [52] was presented to generate additional features of novel classes aided with textual descriptions. To enhance the classification performance of semi-supervised FSL, Zhang et al. [53] devised a semi-supervised meta-generator to produce features around pseudo-labeled samples to calculate more reliable prototypes. Nevertheless, few GAN-based feature generation methods are applied in SAR images. Although Zeng et al. [63] proposed generating the support features to augment the support set, the generated support features were obtained by sampling from the data distribution estimated by transferring the distribution statistics of the base classes rather than using GAN.

Generally, the previous studies [64], [65] are similar to ours. Wang et al. [65], integrated the classifier and the generator into a framework and used the classifier to guide the optimization of the generator. However, the difference is that in [65], the generated images are directly mixed with real images to train the classifier, and no extra module is designed to enhance the quality of the generated images. In [64], a novel diversity transfer network (DTN) was proposed to compensate for the scarce support features. The classifier was also incorporated and trained end-to-end with the generator. DTN focuses on learning the latent diversities between image pairs of training classes and combining them with support features to generate new support features, while our method aims to use GAN to learn the true distribution of support features to generate novel support features.

III. Methodology

Fig. 4 shows the overall framework of the proposed LcFGC. First, the multi-scale feature extraction module of MLC-Net extracts the features of the support samples and query samples. Then, FGC-Net generates and calibrates the support features under the full range of azimuth angles to supplement the real support features. The prototypes for each class are obtained by class-wise averaging the calibrated generated support features and the real support features in the same class. Finally, the prototypes and the embedded query features are fed into the local feature classification module of MLC-Net to predict the class labels of the query samples. The details of the multi-scale feature extraction module, FGC-Net, and the local feature classification module are illustrated in Sections III-A–III-C.

A. Multi-Scale Feature Extraction Module

Previous studies [66], [67] have confirmed the powerful ability of CNNs to extract hierarchical information, where low-level layers extract local detail features contained in images, while high-level layers extract global semantic features. Thus,

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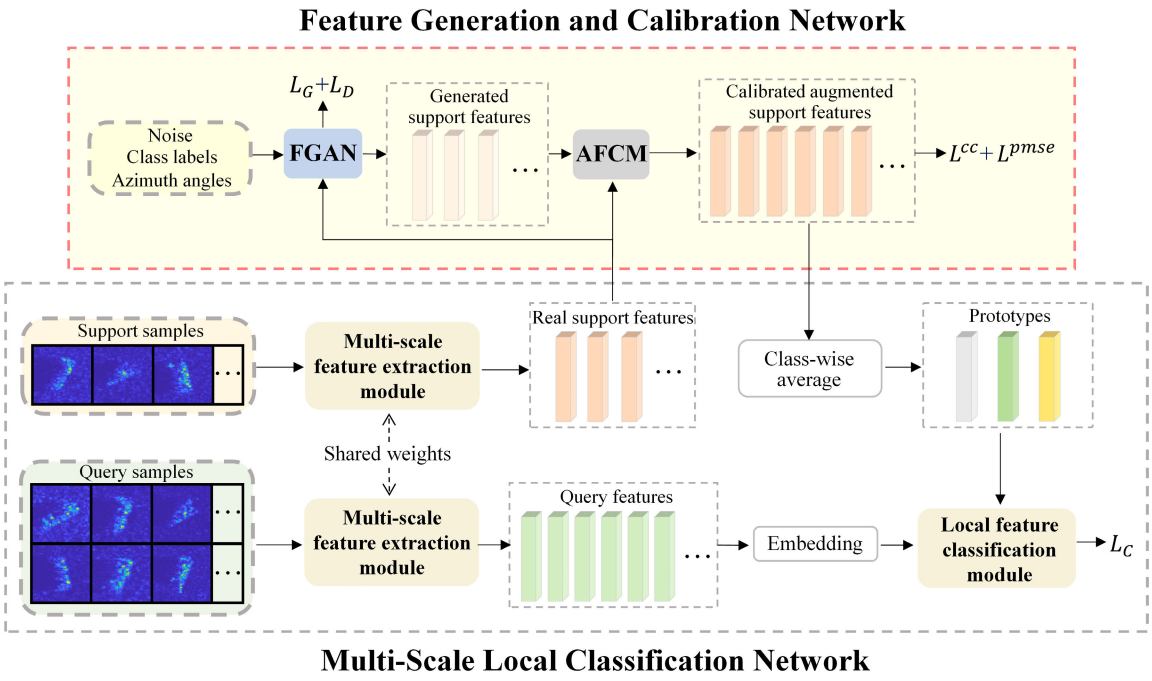


Fig. 4. Overall framework of LcFGC. FGAN and AFCM are the proposed feature GAN and the adaptive feature calibration module, respectively. L_G and L_D are the losses of the generator and discriminator of FGAN. L_C is the classification loss of MLC-Net. L^{cc} and L^{pmse} denote the proposed central constraint loss and the prototype mean square error loss.

multi-scale feature fusion [66], [67], [68], [69], [70] is commonly used to fuse the fine-grained local detail features of low-level layers with the global semantic features extracted by the high-level layers, thus enhancing the discriminability of the target features. Our previous work [70] has demonstrated the effectiveness of multi-scale features in few-shot SAR target classification. Thus, we also employ the multi-scale feature extraction module proposed in [70] in this article.

As shown in Fig. 5, the multi-scale feature extraction module mainly consists of four convolutional blocks, three downsampling blocks, and a transformation block. The four convolutional blocks are employed in the multi-scale feature extraction module to extract multi-scale features gradually. The output features from the first three convolutional blocks are downsampled with the downsampling blocks and fused with the output features from the next convolutional block. Besides, a transformation block filters the redundant information in the multi-scale features to enhance their discriminative ability. Note that the transformation block used in this article does not include the max-pooling operation compared to that in [70] to facilitate the subsequent local feature classification.

The multi-scale feature extraction module introduced above is parameterized as $\varphi_\theta(\cdot)$, where θ denotes its parameters. Given a randomly selected W-way S-shot few-shot classification task, x_i is the ith support sample of class w, $w \in \{1, \ldots, W\}$, and x_j is the jth query sample. Then, the extracted real support features f_r^i of x_i and the extracted query features f_q^j of x_j can be separately denoted as follows:

$$f_r^i = \phi(\varphi_\theta(x_i)) \quad \forall i \in \{1, \ldots, S\} \tag{1}$$

$$f_q^j = \phi\left(\varphi_\theta\left(x_j\right)\right) \quad \forall j \in \{1, \ldots, W \times M\} \tag{2}$$

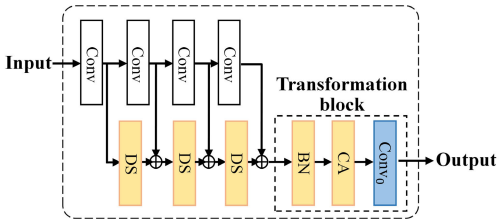


Fig. 5. Network structure of multi-scale feature extraction by referring to [70]. Conv indicates the convolutional block, composed of a convolutional layer, a batch normalization [71] layer, a ReLU nonlinear activation function [72], and a max-pooling layer. DS is the downsampling block and has the same structure as Conv. BN indicates batch normalization, and CA is the operation of channel attention [73]. $\oplus$ means to concatenate the features alone the channel dimension. Compared with Conv, there is no max-pooling operation in Conv$_0$.

where ϕ means to reshape the features $\varphi_\theta(x_i)$ and $\varphi_\theta\left(x_j\right)$ from size $C' \times H' \times W'$ to size $1 \times d$, $d = C' \times H' \times W'$. M indicates the number of query samples per class.

B. Proposed FGC-Net

Due to the limitation of the expensive costs and observation conditions, we usually obtain SAR images under a few azimuth angles in reality. Therefore, the lack of rich information about SAR targets in extensive azimuth angles causes the calculated prototypes to be biased, which affects the classification performance of SAR targets. To overcome this problem, we propose FGC-Net to generate support features of SAR targets at the full range of azimuth angles, which consists of FGAN and AFCM. First, FGAN is devised to generate support features under the full range of azimuth angles, aiming to compensate for the real support features extracted by

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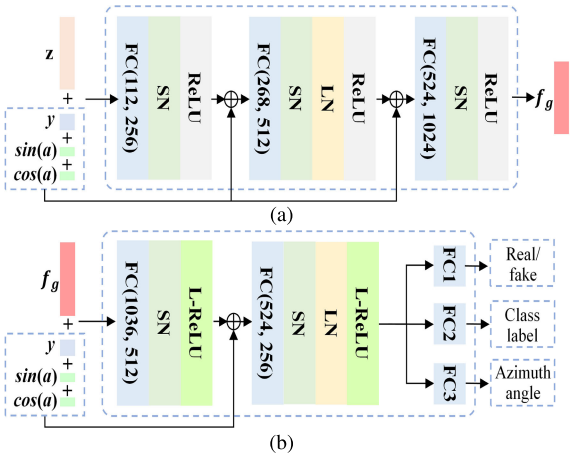


Fig. 6. Network structure of FGAN refers to [48] and [74]. (a) Network structure of the generator of FGAN. (b) Network structure of the discriminator for FGAN. z denotes the noise vector, and y and a are the one-hot class label and azimuth angle, respectively, as conditional information. FC(a, b) represents the fully connected layer with a input nodes and b output nodes. ReLU and L-ReLU are the ReLU activation function and the leakyReLU activation function [75], respectively. SN represents spectral normalization [76], [77], and LN means layer normalization [78]. FC1 indicates the fully connected layer with one output node followed by SN and a sigmoid activation function. FC2 is the fully connected layer with ten output nodes accompanied by SN and a softmax activation function. FC3 has the same structure as FC1 but without the sigmoid activation function. $\oplus$ means to concatenate the conditional information y, $sin(a)$, and $cos(a)$ with the output features.

MLC-Net. Then, we design AFCM to reduce the distribution margin between the generated and real support features. The PMSE loss and CC loss are also integrated to further promote the consistency of the distribution of the generated support features with that of the real support features. The designed FGAN, AFCM, PMSE loss, and CC loss are described below.

1) FGAN: Nowadays, some GAN-based generative models [42], [43], [44], [58], [74] that utilize class labels and azimuth angles as conditional information have also been proposed and applied to the field of SAR images. However, image generation and classification of these above methods are two separate phases, and we do not know whether the generated images are helpful for downstream classification. According to previous feature generation methods [49], [50], [51], [52], [53], we think that feature generation has these advantages over image generation when combined with classification together: 1) generating image features with lower dimensions can be accomplished using a shallower and simpler network, significantly saving computational resources; 2) generating abstract image features with global semantic information is much easier than generating images with detailed information and complex spatial relationships; and 3) features determine the classifier's performance, and generating image features with strong discriminative ability can directly promote the performance of the few-shot classifier. Thus, inspired by existing GAN-based generative models [42], [48], [74], [79], we design FGAN to generate SAR support features at the full range of azimuth angles in each class, where the class label and azimuth angle serve as conditional information, as shown in Fig. 6.

The generation loss L_G of the generator of FGAN is shown as follows:

$$L_G = -E_{f\sim p_g(f)}\left[\log t'\right] + L_G^c + L_G^a \tag{3}$$

$$L_G^c = -E_{f\sim p_g(f)}\left[y \cdot \log y'\right] \tag{4}$$

$$L_G^a = E_{f\sim p_g(f)}\left[\left\|a - a'\right\|_2^2\right] \tag{5}$$

where L_G^c and L_G^a are introduced according to methods [42], [74] to control the generator to generate SAR features at the expected class labels and azimuth angles. y and a are the conditional information concerning the one-hot class label and the azimuth angle. $p_g(f)$ means the distribution of generated features. t', y', and a' are the outputs of the discriminator. Specifically, t' is the probability that the feature f comes from the real feature distribution rather than the generated distribution. y' represents a vector with the same size as y, of which the ith element denotes the probability that f is predicted as class i. a' is the predicted azimuth angle of f.

Likewise, the discrimination loss L_D of the discriminator of FGAN is formulated below

$$L_D = -E_{f\sim p_r(f)}\left[\log t'\right] - E_{f\sim p_g(f)}\left[\log\left(1 - t'\right)\right] + L_D^c + L_D^a \tag{6}$$

$$L_D^c = -E_{f\sim p_r(f)}\left[y \cdot \log y'\right] \tag{7}$$

$$L_D^a = E_{f\sim p_r(f)}\left[\left\|a - a'\right\|_2^2\right] \tag{8}$$

where $p_r(f)$ denotes the real feature distribution. L_D^c and L_D^a are also implemented to enable the discriminator to learn the correct information about class labels and azimuth angles for the real features.

Following the FGAN raised above, we can generate support features at the full range of azimuth angles. Although generating support features within the same range of azimuth angles as the test data will benefit the classification, it is not achievable since we do not know the azimuth angles of the test data. Therefore, we generate support features at the full range of azimuth angles to maximize the classification accuracy. Concretely, for the public MSTAR dataset that we are interested in, the azimuth angles of SAR images are approximately uniformly distributed at [0°, 360°) with an interval of about 5°. Moreover, it has been verified earlier in [80] and [81] that the scattering centers of SAR targets can remain stable at small azimuth intervals (i.e., [−5°, 5°]). Thus, we adopt a 5° azimuth interval to generate the support features, aiming to match the distribution of the MSTAR data while saving computational resources. Given the ith real support feature f_r^i of class w with an azimuth angle a_r^i, the generated support features for f_r^i are represented as $\{(f_g^{it}, w, a_r^i + 5*t), t = 1, \ldots, T\}$, where $T = 360/5 = 72$.

2) AFCM: For few-shot SAR target classification, there are only a few real images under certain azimuth angles for each class of target domain. Therefore, FGAN fails to effectively capture the information of the targets in the full range of azimuth angles, which causes the distribution differences between the generated features and the real ones under the specified azimuth angles. Hence, inspired by the self-attention mechanism [54], AFCM is presented to adaptively re-weight the real and generated support features of the same class,

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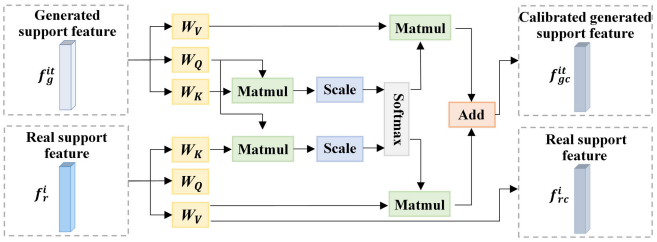


Fig. 7. Operations of AFCM. Take the example of each class containing only one real support sample. Matmul indicates the matrix multiplication, Scale means to scale the input feature, and Add denotes the sum operation of the two input features.

intending to calibrate the generated support features according to the real support features.

Specifically, we take the tth support feature $f_g^{it} \in R^{1\times d}$ generated for the ith real support feature f_r^i of class w as an example to illustrate the calibration process. The generated support feature f_g^{it} of the wth class is calibrated with all the real support features $\{f_r^i, i = 1, \ldots, S\}$ from the same class. As displayed in Fig. 7, the weights $W_K \in R^{d\times d}$, $W_Q \in R^{d\times d}$, and $W_V \in R^{d\times d}$ are first employed to linearly map the generated support feature f_g^{it} and the real support features $\{f_r^i, i = 1, \ldots, S\}$ into a new feature space to obtain the respective keys $K \in R^{(S+1)\times d}$, queries $Q \in R^{(S+1)\times d}$, and values $V \in R^{(S+1)\times d}$

$$\begin{cases} K = FW_K \\ Q = FW_Q \\ V = FW_V \end{cases} \tag{9}$$

where $F = [f_g^{it}; f_r^1; f_r^2; \ldots; f_r^S]$, $F \in R^{(S+1)\times d}$.

Then, the calibrated generated support feature f_{gc}^{it} of f_g^{it} is computed as a weighted sum of the values V. The corresponding coefficients are obtained via the inner product between the query of f_g^{it} and the keys K. The calibrated generated support feature f_{gc}^{it} is calculated as

$$f_{gc}^{it} = \text{softmax}\left(\frac{q_1 K^T}{\sqrt{d}}\right) V \tag{10}$$

where $q_1 \in R^{1\times d}$ is the first row in the matrix Q that represents the query of f_g^{it}.

Finally, we can obtain the calibrated augmented support feature set $F_{S_w} = F_{S_w}^r \cup F_{S_w}^g$ of class w to calculate the prototype, where $F_{S_w}^r = \{(f_{\text{rc}}^i, w, a_r^i), i = 1, \ldots, S\}$ is the real support feature set of class w, and f_{rc}^i is the value of f_r^i mapped by the weights W_V as described in Fig. 7. $F_{S_w}^g = \{(f_{gc}^{it}, w, a_r^i + 5 \times t)), i = 1, \ldots, S, t = 1, \ldots, T\}$ denotes the calibrated generated support feature set of class w.

3) PMSE Loss and CC Loss: To further align the distribution of the calibrated generated support features with the real support features, we also present the PMSE loss and CC loss in this section.

a) PMSE Loss: First, we design the PMSE loss to restrain the consistency of the global distribution of the real support features and the calibrated generated support features, which is achieved by simply constraining the mean square error (MSE) loss between the prototypes of both the real support features and the calibrated generated support features in the same class. The PMSE loss on the W-way S-shot few-shot classification task is expressed as

$$L^{\text{pmse}} = \frac{1}{W} \times \left(\sum_{w=1}^{W} \text{MSE}\left(m_r^w, m_g^w\right)\right) \tag{11}$$

where MSE denotes the MSE loss function. m_r^w and m_g^w are the prototypes according to the real support features $\{f_{\text{rc}}^i, i = 1, \ldots, S\}$ of the wth class and the calibrated generated support features $\{f_{gc}^{it}, i = 1, \ldots, S, t = 1, \ldots, T\}$ within class w, respectively, which are calculated as follows:

$$m_r^w = \frac{1}{S} \times \sum_{i=1}^{S} f_{\text{rc}}^i \tag{12}$$

$$m_g^w = \frac{1}{S \times T} \times \sum_{i=1}^{S} \sum_{t=1}^{T} f_{gc}^{it}. \tag{13}$$

b) CC Loss: As described in Section III-B2), for f_{rc}^i, T calibrated generated support features $\{f_{gc}^{it}, t = 1, \ldots, T\}$ exist. The azimuth angles of these $T + 1$ support features are theoretically distributed equally over 0°–360°, and we thus name the $T + 1$ support features as the ith cluster. For a W-way S-shot few-shot classification task, S clusters of the same class should be close to each other. Hence, the CC loss is devised to constrain the S clusters from the local point of view. The center of the ith cluster within class w is calculated as follows:

$$c_{wi} = \frac{1}{T+1} \times \left(f_{\text{rc}}^i + \sum_{t=1}^{T} f_{gc}^{it}\right). \tag{14}$$

Afterward, S centers of the same class are constrained by MSE loss to drive S clusters close to each other in the feature space. The CC loss over a few-shot classification task is calculated below

$$L^{\text{cc}} = \frac{1}{W} \times \left(\sum_{w=1}^{W} \sum_{m<n} \text{MSE}(c_{wm}, c_{wn})\right) \tag{15}$$

where $m \in \{1, \ldots, S-1\}$ and $n \in \{2, \ldots, S\}$.

C. Local Feature Classification Module

As depicted in Fig. 1, for SAR images with large inter-class similarity, the contribution of critical local features of targets in SAR images to the classification is weakened using the metric-based meta-learning approach to calculate the similarity between global features. Therefore, to enhance the assistance of local detail features of SAR targets for classification, we propose a local feature classification module following the approach of [38] to refine prototypes and query features into many local region features (LRFs) for classification, as shown in Fig. 8.

Concretely, with a W-way S-shot few-shot classification task, we can obtain the calibrated augmented support feature set F_{S_w} of class w and the query feature set F_Q following the abovementioned methods. The prototype $m_w \in R^{1\times d}$ of class w is calculated as

$$m_w = \frac{1}{S \times (1+T)} \times \sum_{i=1}^{S} \left(f_{rc}^i + \sum_{t=1}^{T} f_{\text{gc}}^{it}\right). \tag{16}$$

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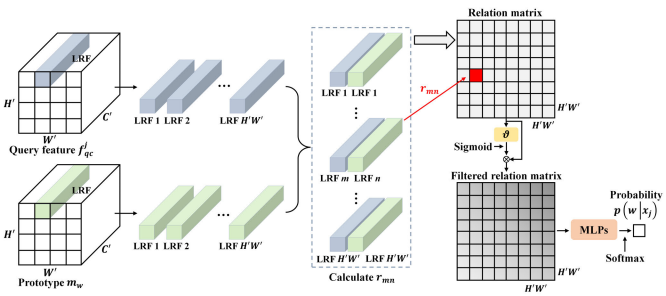


Fig. 8. Classification process of local feature classification module, where ϑ represents the convolutional operation, and $\otimes$ means the element-wise multiplication. MLPs denotes the fully connected layers.

Then, for the query feature $f_q^j \in F_Q$ of the jth query sample x_j, we first perform an embedding with weights W_V to ensure the consistency of the feature space

$$f_{\mathrm{qc}}^j = f_q^j W_V. \tag{17}$$

After that, the dimensions of the query feature f_{qc}^j and the prototype of each class are reshaped from $1 \times d$ to $C' \times H' \times W'$, aiming to recover the original spatial relationship of the target. In the following, we refine the query feature f_{qc}^j and the prototype of each class into several LRFs for predicting the class label of x_j.

As depicted in Fig. 8, the query feature f_{qc}^j and prototype m_w can be regarded as a combination of $H'W'$ C'-dimensional LRFs. Thus, f_{qc}^j and m_w can be represented as $\{l_j^m, m = 1, \ldots, H'W'\}$ and $\{l_{m_w}^n, n = 1, \ldots, H'W'\}$, respectively, where l_j^m is the mth LRF of f_{qc}^j and $l_{m_w}^n$ is the nth LRF of m_w.

The class label of query sample x_j is predicted by evaluating the relation matrix of its query feature f_{qc}^j and the prototype of each class. In particular, the relation matrix about f_{qc}^j and m_w is represented as $\mathrm{RM}_{jw} = (r_{mn})_{H'W' \times H'W'}$. r_{mn} represents the intersection element located in the mth row and nth column of RM_{jw}, which indicates the semantic similarity between the mth LRF of f_{qc}^j and the nth LRF of m_w. The calculation of r_{mn} is given below

$$r_{mn} = -d\left(l_j^m, l_{m_w}^n\right) \quad \forall m, n \in \left\{1, \ldots, H'W'\right\} \tag{18}$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance. The smaller the Euclidean distance, the higher the semantic similarity between the features, and vice versa.

After the semantic similarity among all LRFs of f_{qc}^j and m_w is calculated, the convolutional operation is performed to enhance the similarities between local detail features that greatly assist classification and suppress those not necessary for classification. The filtered relation matrix RM_{jw}^f is expressed as follows:

$$\mathrm{RM}_{jw}^f = \sigma\left(\vartheta\left(\mathrm{RM}_{jw}\right)\right) \otimes \mathrm{RM}_{jw} \tag{19}$$

where ϑ represents a convolutional operation, σ is the sigmoid activation function, and $\otimes$ means the element-wise multiplication.

RM_{jw}^f is then input into the fully connected layers, and thus we can get the final similarity s_{jw} between the query feature f_{qc}^j and the prototype m_w

$$s_{jw} = \mathrm{MLPs}\left(\tilde{\phi}\left(\mathrm{RM}_{jw}^f\right)\right) \tag{20}$$

where $\tilde{\phi}$ means to reshape RM_{jw}^f from size $H'W' \times H'W'$ to size $1 \times D$, $D = H'W' \times H'W'$, MLPs indicates two fully connected layers whose first layer is followed by a ReLU nonlinear activation function.

Finally, the softmax function is applied to get the probability of query sample x_j belongs to class w

$$p\left(w \mid x_j\right) = \frac{\exp\left(s_{jw}\right)}{\sum_{w'} \exp\left(s_{jw'}\right)}. \tag{21}$$

With the probabilities obtained above, we can calculate the classification loss over a W-way S-shot few-shot classification task

$$L_C = -\frac{1}{W \times M} \sum_{j=1}^{W \times M} \sum_{w=1}^{W} I\left(y_j = w\right) \log p\left(w \mid x_j\right) \tag{22}$$

Algorithm 1 Training Algorithm of LcFGC in the W-Way S-Shot Case

Input: Training set D_{train} in the source domain; selected support set D_{sup} in the target domain; rotated support set D_{sup}^r, which is obtained by randomly rotating each sample in D_{sup} three times and merging them with D_{sup}; $D_{all} = D_{train} \cup D_{sup}^r$.
Output: The well-trained LcFGC.

for $i = 1 : n$ **do**
  Optimization of generator:
  for $j = 1 : 3$ **do** (Generator is optimized three times)
    Randomly select a W-way S-shot few-shot classification task T_i^j from D_{all}.
    Send T_i^j into the multi-scale feature extraction module to extract the support features and query features by (1) and (2).
    Using FGAN to generate support features.
    Calculate the loss of the generator by (3)-(5).
    Calibrate the generated support features through AFCM.
    Calculate the PMSE loss and CC loss by (11)-(15).
    Send the calibrated generated and real support features and query features to the local feature classification module to calculate the classification loss by (16)-(22).
    Update the parameters of the generator by (23).
  end for
  Optimization of discriminator:
  Repeat steps 3 to 5.
  Calculate the loss of the discriminator by (6)-(8).
  Update the parameters of the discriminator by (6).
  Optimization of AFCM and MLC-Net:
  Repeat steps 3 to 9.
  Update the parameters of AFCM and MLC-Net by (24).
end for

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TABLE I
SAMPLE NUMBERS OF TEN CLASSES OF TARGETS IN THE MSTAR DATASET

Target	BMP2			BTR70	T72			BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU23/4
Class	SNC21	SN9563	SN9566	C71	SN132	SN812	SNS7	k10yt7532	b01	E-71	92v13015	A51	E12	d08
17°	0	233	0	233	232	0	0	256	299	298	299	299	299	299
15°	196	195	196	196	196	195	191	195	274	274	274	273	274	274

where M is the number of query samples per class. $I(\cdot)$ is the indicator function. $I(\cdot)$ outputs 1 when the true class label y_j of x_j equals w and outputs 0 otherwise.

D. Optimization of LcFGC

During the training process, the optimization of LcFGC in an iteration is divided into three parts: optimization of the generator of FGAN, optimization of the discriminator of FGAN, and optimization of AFCM and MLC-Net. At first, we optimize the generator and discriminator of FGAN sequentially. After that, the generated and real support features are fed into AFCM and MLC-Net successively for feature calibration and classification. Then we update the parameters of AFCM and MLC-Net. The details about the training process of LcFGC are shown in Algorithm 1.

1) Optimization of Generator: To enable MLC-Net to guide FGAN to generate support features favorable for classification directly, we first combine the classification loss L_C of MLC-Net with the generation loss L_G of the generator. Meanwhile, the PMSE loss L^{pmse} and the CC loss L^{cc} are added, considering the alignment of the distribution of the generated support features with the real ones. Accordingly, the total loss function of the generator is presented below

$$L_G^{\text{total}} = L_G + \lambda \cdot \left(L_C + L^{\text{pmse}} + L^{\text{cc}}\right) \tag{23}$$

where λ is a hyperparameter, and the value of λ is set to 5 after sensitivity analysis.

2) Optimization of Discriminator: The loss function for the discriminator of FGAN is kept the same as (6).

3) Optimization of AFCM and MLC-Net: Considering that MLC-Net extracts the real support features, we attach the PMSE loss L^{pmse} and CC loss L^{cc} to the classification loss L_C of MLC-Net, to constrain the distribution of the real support features to be closer to the distribution of the generated support features. Hence, the total loss function is shown in the following:

$$L_C^{\text{total}} = L_C + \beta \cdot \left(L^{\text{pmse}} + L^{\text{cc}}\right) \tag{24}$$

where β is a hyperparameter, and the value of β is set to 1 after sensitivity analysis.

IV. EXPERIMENTS AND RESULTS

A. Introduction of MSTAR Dataset

The experiments in this article are performed on the MSTAR dataset provided by the Defense Advanced Research Projects Agency. The images were collected by an SAR sensor with a spatial resolution of 0.3 × 0.3 m in range and azimuth directions. The optical photographs of ten classes of targets in the MSTAR dataset are shown in Fig. 9, and the sample numbers of each class are displayed in Table I.



Fig. 9. Optical photographs of ten classes of targets in the MSTAR dataset.

TABLE II
TRAINING SET OF SOURCE DOMAIN

Target Class	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU23/4
Training set	256	299	298	299	299	299	299

TABLE III
SUPPORT SET AND TEST SET OF TARGET DOMAIN

Target	BMP2			BTR70	T72		
Class	SNC21	SN9563	SN9566	C71	SN132	SN812	SNS7
Support set	0	233	0	233	232	0	0
Test set	196	195	196	196	196	195	191

To simulate the few-shot SAR target classification, the MSTAR dataset is divided into the training set, support set, and test set following the setup in [24] and [70]. Table II shows the target classes and the number of samples for the training set D_{train}. The support set D_s and test set D_{test} are described in Table III, where the test set contains the variants for the targets of the support set to increase the complexity of the few-shot SAR target classification.

We crop the images of the MSTAR dataset from size 128 × 128 to 64 × 64 to eliminate the influence of background in SAR images on classification. The cropped SAR images are then processed by L2 normalization and sent to LcFGC.

B. Details of Implementation

In the multi-scale feature extraction module, the kernel size of the convolutional block is set to 3 × 3 according to the typical setting in [36]. The kernel size for the convolutional operation shown in (19) is 7 × 7, and the parameter "padding" is set to "SAME" to ensure that the output and

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TABLE IV
COMPARISON OF THE CLASSIFICATION PERFORMANCE IN THE 3-WAY S-SHOT CASE

Method	5-shot	10-shot	20-shot
A-ConvNet [1]	57.51%	68.04%	78.82%
DA-Net [20]	58.47%	65.13%	73.50%
ARGN [24]	53.49%	67.44%	82.73%
MSAR [34]	75.76%	78.57%	82.08%
PN [36]	75.12%	78.37%	81.08%
RN [37]	72.54%	76.66%	77.88%
Meta-Baseline [84]	77.18%	79.41%	83.25%
AG-MsPN [70]	82.03%	87.57%	92.68%
Ours	**85.33**%	**91.32**%	**94.53**%

TABLE V
EFFECTIVENESS OF THE COMPONENTS OF LCFGC ON CLASSIFICATION

Baseline	LFCM	FGC-Net	5-shot	10-shot	20-shot
✓			79.36%	85.18%	91.14%
✓	✓		81.15%	85.35%	91.97%
✓	✓	✓	**85.33**%	**91.32**%	**94.53**%

TABLE VI
ABLATION STUDY ON FGC-NET

FGAN	AFCM	PMSE	CC	5-shot	10-shot	20-shot
				81.15%	85.35%	91.97%
✓				75.80%	80.65%	86.73%
✓	✓			80.27%	84.98%	88.81%
✓	✓	✓		84.70%	90.44%	93.85%
✓	✓	✓	✓	**85.33**%	**91.32**%	**94.53**%

input dimensions are the same. In (20), the output node of the last fully connected layer is set to 1, which indicates the correlation between the query sample and the prototype of a particular class. The output node of the first fully connected layer in (20) is set to 64.

Before training LcFGC, we randomly select S support samples for each class from the support set, as illustrated in Table III, to simulate the 3-way S-shot situation, where the values of S are assigned to 5, 10, and 20. Then, each sample in the selected support set D_{sup} is randomly rotated three times and then merged with D_{sup} to obtain D^r_{sup}.

1) Pretraining of MLC-Net: To improve the credibility of the real support features extracted by MLC-Net at the early stage of training, we first train MLC-Net for 2000 episodes using D_{train} and D^r_{sup}. Specifically, an ADAM optimizer [82] with an initial learning rate of 0.001 is utilized, and the learning rate is annealed by 1/2 for every 400 episodes.

2) End-to-End Training of FGC-Net and MLC-Net: The pre-trained MLC-Net is combined with FGC-Net for end-to-end training of 10 000 episodes using D_{train} and D^r_{sup}, as illustrated in Algorithm 1. In the end-to-end training stage, the ADAM optimizer is used to update all parameters of LcFGC. The initial learning rates of the generator and discriminator are set to 0.0001 and 0.00001, respectively. As for MLC-Net, the learning rate starts from 0.001 and decays to 1/2 of the previous ones every 5000 episodes. The general technique of gradient clipping by global norm [83] with clipping ratio 1.0 is introduced into the optimization of MLC-Net, aiming to avoid unstable training caused by FGAN.

C. Results

To adequately demonstrate the superiority of the proposed LcFGC, various approaches are compared: data augmentation-based [1], [20], transfer learning-based [24], optimization-based meta-learning [34], and metric-based meta-learning [36], [37], [70], [84]. We trained each method 20 times and reselected D_{sup} every time to eliminate the randomness of the classification results introduced by the selection of the support set D_{sup}.

The comparison of the average classification performance of LcFGC with the other eight methods is shown in Table IV, where the classification results of the eight comparison methods were reported by Wang et al. [70]. As illustrated in Table IV, the performance of the proposed LcFGC is the highest in 5-, 10-, and 20-shot scenarios. Table IV shows that the performance of A-ConvNet and DA-Net based on the data augmentation is much inferior to that of LcFGC, which indicates that data augmentation cannot fundamentally solve the SAR target classification in the case of extreme scarcity of SAR data. With the aid of the angular-related transferable knowledge learned from the SAR data in the source domain, the performance of ARGN is comparable to the meta-learning methods (i.e., MSAR, PN, and Meta-Baseline) in the case of 20-shot. However, in the severe 5- and 10-shot cases, the generalization performance of ARGN on the unseen classes of the target domain is worse than the above methods because no episodic training strategy is applied. Compared to the metric-based meta-learning methods PN, RN, Meta-Baseline, and AG-MsPN, our method outperforms them by at least 3.30%, 3.75%, and 1.85% in classification accuracy in 5-, 10-, and 20-shot scenarios. This phenomenon verifies that the support features generated by the designed FGC-Net under the full range of azimuth angles compensate for the real support features, resulting in more reliable prototypes. Moreover, with the support of the local feature classification module, the critical local detail features on SAR images are emphasized, thereby enhancing the performance of the few-shot SAR target classification.

D. Ablation Study

We further implement ablation experiments in this section to validate the importance of the proposed local feature classification module (LFCM) and FGC-Net for few-shot SAR target classification, respectively. All the experiments are conducted on the training, support, and test sets shown in Tables II and III.

1) Effectiveness of LFCM and FGC-Net: We performed a series of ablation experiments to analyze the contribution of

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LFCM and FGC-Net to the few-shot SAR target classification. The average results of 20 times of experiments with different methods are presented in Table V. First, we use the multi-scale feature extraction module followed by a non-parametric nearest-prototype classification method [36] as the baseline. That is, the multi-scale feature extraction module extracts the features of support and query images. Then, the prototypes of each class are obtained by taking the mean value of the support features of the same class. Finally, the class label of each query sample is predicted by finding the prototype with the minimum Euclidean distance from its query feature.

As shown in Table V, with the addition of LFCM, the classification accuracy of the baseline is improved to different degrees in the 5-, 10-, and 20-shot scenarios, respectively, which confirms that the proposed LFCM is better suited for SAR image classification rather than directly calculating the similarity between the global features and prototypes. FGC-Net is integrated into MLC-Net to generate SAR support features at the full range of azimuth angles to correct the biased prototypes. Upon further incorporating the designed FGC-Net, the classification performance of MLC-Net (i.e., the baseline model with LFCM) is improved by approximately 4%, 5%, and 2% for the 5-, 10-, and 20-shot scenarios, respectively. Since the designed FGC-Net can generate support features for each class of SAR targets under the full range of azimuth angles, the prototypes are corrected by the generated support features together with the real support features, thus improving the performance of few-shot SAR target classification.

2) Ablation Analysis of FGC-Net: The presented FGC-Net includes FGAN and AFCM. PMSE and CC loss are also integrated to facilitate the alignment of the calibrated generated support features to the real support features. In this section, FGAN, AFCM, PMSE loss, and CC loss are sequentially attached to MLC-Net to explore the importance of the above components. The mean values of the results of 20 times of experiments are reported in Table VI.

The first row of Table VI shows the classification performance of MLC-Net (i.e., the baseline with LFCM in Table V). As shown in Table VI, the classification performance degrades in all scenarios when directly integrating FGAN into MLC-Net. The possible reason is that the difference in distribution between the generated and real support features leads to performance degradation. Hence, we introduce AFCM, which is dedicated to calibrating the generated support features with the real support features of the same class. As shown in the experimental results in the third row of Table VI, adding AFCM improves the network's classification accuracy by about 4%, 4%, and 2% for the 5-, 10-, and 20-shot cases, respectively, aptly demonstrating the effectiveness of AFCM. And yet, at this time, the classification performance after adding FGAN and AFCM to MLC-Net still does not exceed that of MLC-Net, indicating that the distribution of currently generated support features is not yet in excellent alignment with that of the real support features. Therefore, we further design the PMSE loss and CC loss, hoping to achieve the alignment between the distribution of the generated support features and the real support features under the constraint of both. As can be seen from the experimental results in the last two rows of Table VI, with the help of PMSE loss and CC loss, the distribution difference between the generated and real support features is further mitigated, resulting in a remarkable enhancement in the performance of few-shot SAR target classification.

3) Validation of Generative Capability of FGC-Net: To compensate for the degradation of classification performance caused by the absence of SAR target images at numerous azimuth angles, we design FGC-Net to generate SAR support features at the full range of azimuth angles for each class of targets. The experimental results presented in Table VI have quantitatively analyzed the effectiveness of FGC-Net. Due to the complexity of generating support features under the full range of azimuth angles, we analyze the generative capability of FGC-Net in terms of the azimuth angles in this section. The models in the second and last rows of Table VI are simplified as MLC-Net-FGAN and MLC-Net-FGC-Net.

To evaluate the precision of the azimuth angles of the support features generated by FGC-Net, the real support features of the T72 target in the target domain and the corresponding generated support features are visualized using t-SNE [85] according to azimuth angles. Specifically, in the 3-way S-shot few-shot classification task, there exist S randomly selected real support features of the T72 target whose azimuth angles are randomly distributed from 0° to 360°, where $S = 5, 10, 20$. We generate T support features for each real support feature of the T72 target according to the method shown in Section III-B, where the azimuth angles of the generated T support features are uniformly distributed from 0° to 360° with an interval of 5°.

Fig. 10 displays the t-SNE visualization of S real support features and $S \times T$ generated support features of the T72 target extracted by MLC-Net-FGAN and MLC-Net-FGC-Net in the 5-, 10-, and 20-shot scenarios. As shown in Fig. 10(a)–(c), under MLC-Net-FGAN, the support features generated by FGAN under the full range of azimuth angles are distributed according to the azimuth angles. But at this time, the real support features at different azimuth angles are clustered instead of distributed according to azimuth angles. It indicates that the distribution of the generated and real support features is inconsistent. From Fig. 10(d)–(f), we can see that most of the real support features and the generated support features whose azimuth angles belong to the same range are close to each other in the feature space in the 10- and 20-shot cases. Due to the extreme sparsity of the real support samples provided in the 5-shot case, the alignment between the generated and real support features is not as satisfactory as that in the 10-shot and 20-shot cases. Considering all the visualization images together, the designed FGC-Net indeed promotes the consistency of the azimuth angles of the generated and real support features.

4) Verification of Prototype Calibration: In this article, we propose FGC-Net to generate SAR support features at all ranges of azimuth angles to compensate for the real support features and correct the bias of the prototype. In the following, we will verify that the calculated prototype is indeed

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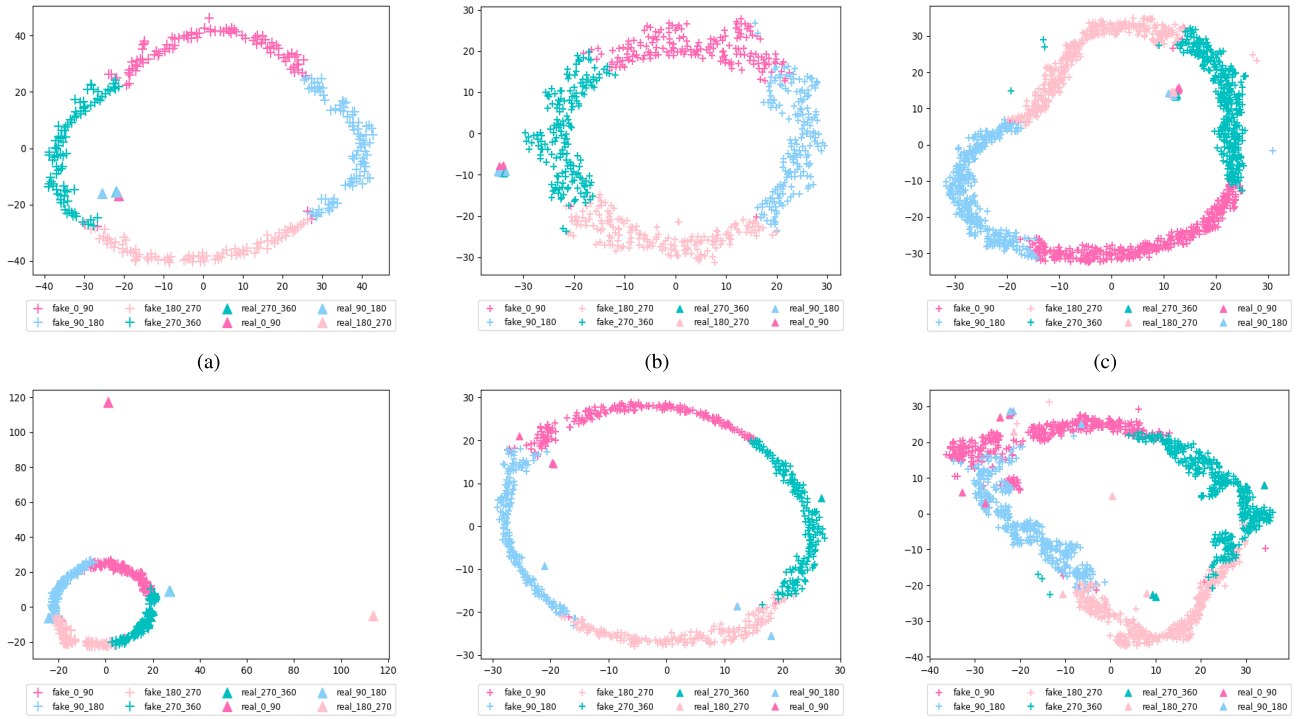


Fig. 10. Visualization of real support features and the corresponding generated support features of the T72 target extracted by (a)–(c) MLC-Net-FGAN and (d)–(f) MLC-Net-FGC-Net under 5-, 10-, and 20-shot cases. The features whose azimuth angles lie in the same interval are marked with the same color. fake_i_j represents the generated support features with azimuth angle a_g, $a_g \in [i^\circ, j^\circ)$. Similarly, real_$i$_$j$ represents the real support features with azimuth angle a_r, $a_r \in [i^\circ, j^\circ)$.

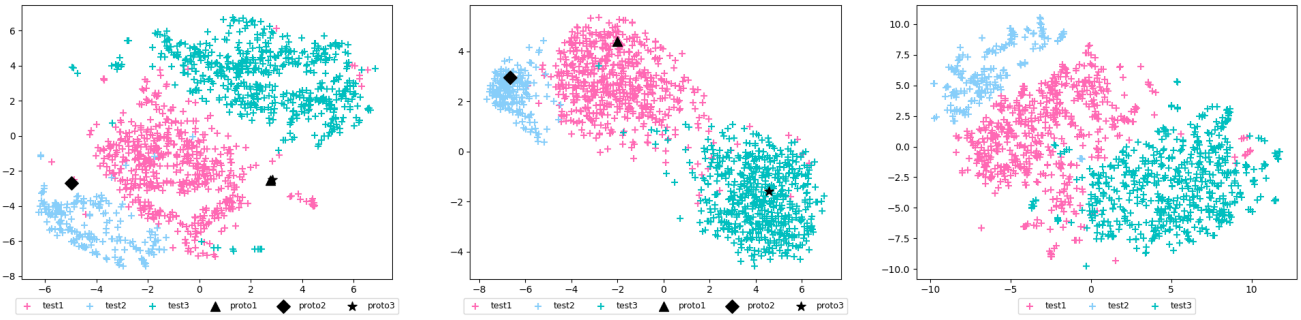


Fig. 11. Verification of prototype calibration. (a) t-SNE visualization of the prototypes and features of the three classes of test data extracted by MLC-Net. (b) t-SNE visualization of the prototypes and features of the three classes of test data extracted by LcFGC. (c) t-SNE visualization of the features of the three classes of test data extracted by PN. Test$i$ means the features of the test data of class i, and proto$i$ represents the prototype of class i.

calibrated after adding FGC-Net to the basic few-shot classifier MLC-Net.

Specifically, we train MLC-Net and LcFGC using the training set and support set shown in Tables II and III for the 20-shot scenario, respectively. Then, the test features and corresponding prototypes of the three classes in the target domain extracted by the trained MLC-Net and LcFGC are visualized in Fig. 11(a) and (b), respectively. As illustrated in Fig. 11(a), only the prototype and the test features of the second class are closer in the feature space. In both the first and third classes, there is a significant drift in the distribution of prototype and test features. This indicates that the prototypes obtained using only a few real support features are biased. However, when FGC-Net with PMSE loss and CC loss was introduced on MLC-Net, the distribution of the prototypes became consistent with the corresponding class of test features, which directly verified that the accurate support features under the full range of azimuth angles generated by FGC-Net did eliminate the bias of the prototype, thus improving the performance of few-shot SAR target classification.

In addition, the high inter-class similarity and intra-class variability present in SAR images also affect the few-shot classification performance. In this section, we also incorporate t-SNE visualizations to validate LcFGC's ability for handling

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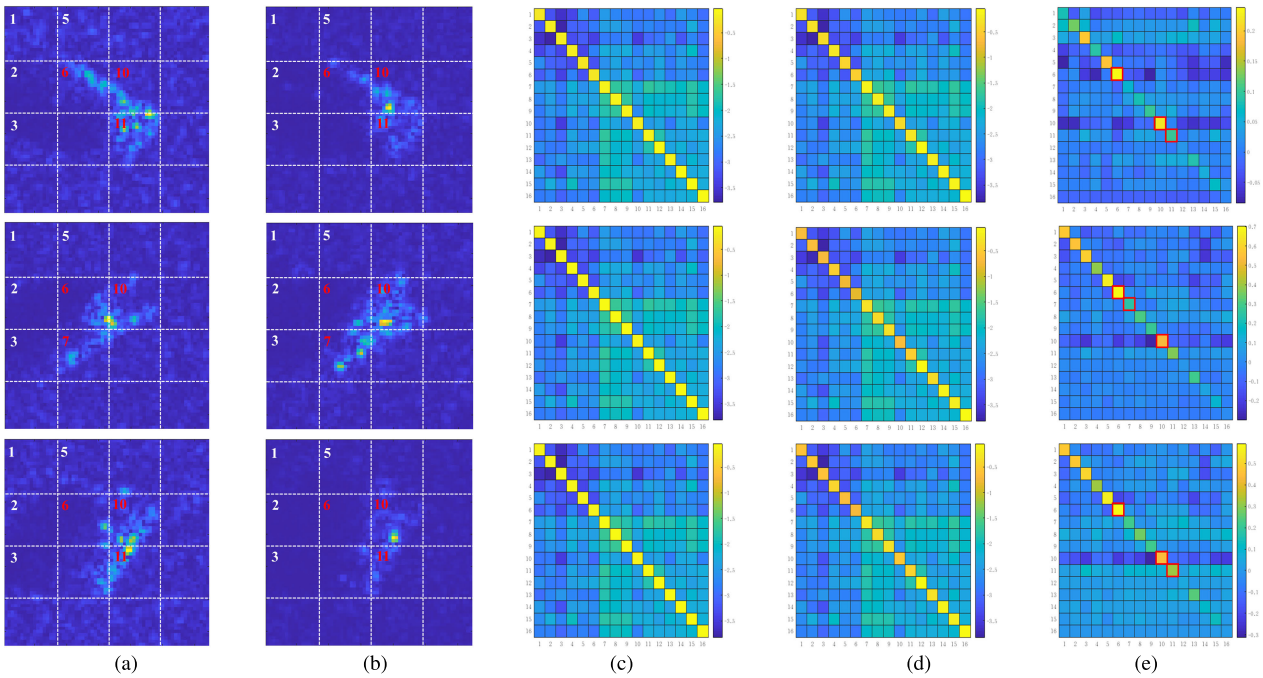


Fig. 12. Three pairs of test SAR images of BMP2 and T72 targets, the FRMs between their image features and the prototype of BMP2, and the FRM difference maps, where the test SAR images of BMP2 and T72 targets are correctly classified. (a) BMP2. (b) T72. (c) FRMs of BMP2. (d) FRMs of T72. (e) FRM difference maps.

this problem briefly. Fig. 11(c) visualizes the test features extracted by PN. Combining Fig. 11(a)–(c), it is noticed that by adding the multi-scale feature extraction module, LFCM, and FGC-Net sequentially into PN, the test features of the same class become compactly clustered together, and the boundaries between the features of different classes become pretty distinct. This phenomenon validates that the multi-scale feature extraction module, LFCM, and FGC-Net in the LcFGC can gradually deal with the problem of both inter-class similarity and the intra-class variability of SAR images.

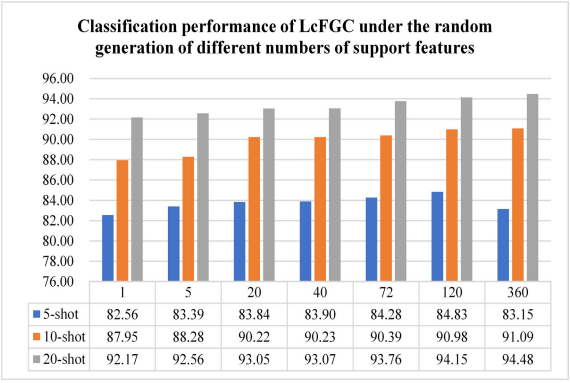
E. Influence of Local Detail Features on Classification

In this article, we employ LFCM to enhance the contribution of local detail features to the classification. To assess the significance of the SAR target's local detail features in classification, we analyze the difference maps of the filtered relation matrix (FRM) between the test image features and the prototypes in the 20-shot case.

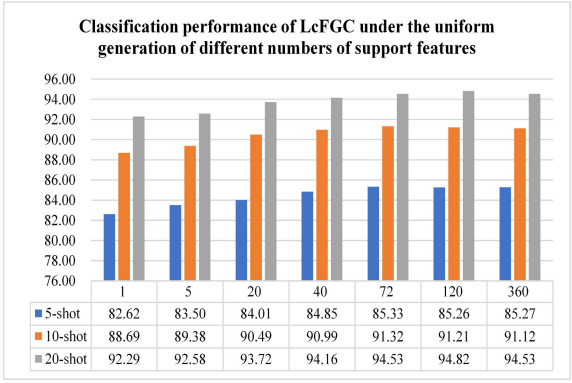
Specifically, we first divide each test image feature and prototype into 16 local region features (LRFs). Then, according to (18) and (19), the FRM with a size of 16×16 between the test image feature and the prototype can be calculated. Finally, the difference map is obtained by calculating the difference between any two FRMs. Fig. 12 displays the test SAR images of BMP2 and T72 targets, the FRMs between their image features and the prototype of BMP2, and the FRM difference maps, in which the test SAR images of BMP2 and T72 targets are correctly classified.

By combining Fig. 12(c)–(d), it can be observed that there are strong correlations between the LRFs of SAR images and the LRFs of the prototypes on the diagonal in FRM. This correlation exists regardless of whether the categories of SAR images and prototypes are consistent. This indicates that the proposed LFCM forces the LRF of the SAR image to be most similar to the LRF of the prototype at the same position. In simpler terms, LFCM constrains the LRF of the SAR image in the target region to be closest to the LRF of the prototype in the corresponding target region rather than the LRF of the background region. Hence, in Fig. 12(d), the values of FRMs between the T72 image features and the prototype of BMP2 are still maximum on the diagonal. When both BMP2 and T72 targets are correctly classified, it can be observed from Fig. 12(e) that there are only a few diagonal elements with significant differences in the FRM difference maps, indicating that only a limited number of local regions play a key role in distinguishing the two targets. For instance, as shown in the third row of Fig. 12, the SAR images of BMP2 and T72 targets have noticeable visual differences at the sixth, tenth, and 11th local positions. Meanwhile, their FRM difference maps have large values at the coordinate positions (6, 6), (10, 10), and (11, 11), properly validating the ability of the proposed LFCM to capture the subtle differences between similar targets. In addition, the FRM difference maps between BMP2 and T72 also have significant differences in the local positions corresponding to the background areas (e.g., locations 2 and 3). This is because locations 2 and 3 correspond to the shadow of the SAR target, which reflects the target's contour information. Similarly, locations 1 and 5 reflect the edges of the target or target's shadows, and thus the elements of FRM difference maps at (1, 1) and (5, 5) positions also have higher values. Hence, the proposed LFCM can not only learn the visual disparities between SAR targets but also capture the differences in the targets' shadow regions, thus facilitating the recognition of similar SAR targets.

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Case 1



Case 2

Fig. 13. Influence of the number of generated support features on the classification performance.

Based on the above analysis, we know that the proposed method utilizes FRMs extracted by LFCM to capture the visual disparities of the targets. When comparing two targets with different local visual features, it is noticeable that there are significant differences in the corresponding local features in their FRM. Thus, the visual differences in the original SAR images manifest themselves in the network through the FRM, which shows significant differences at the corresponding local regions for two different targets. In summary, the visual disparities are pivotal in few-shot SAR target classification.

F. Effect of Number of Generated Support Features on LcFGC

In this section, the following two cases of experiments are mainly considered to verify the effect of the number of generated support features on the prototype calibration:

Case 1: For each real support feature, we respectively generate 1, 5, 20, 40, 72, 120, and 360 support features, of which the azimuth angles are randomly sampled;

Case 2: For each real support feature, we respectively generate 1, 5, 20, 40, 72, 120, and 360 support features. Unlike in case 1, the azimuth angles of the generated support features are uniformly distributed from 0° to 360°.

All experiments were performed with 20 different support sets. The average performance of 20 times of experiments for the above cases 1 and 2 are given in Fig. 13.

As shown in Fig. 13, when the number of support features generated for each real support feature is less than or equal to 72, the classification performance of the LcFGC we proposed is steadily improved as the number of generated support features increases in cases 1 and 2, which indicates that generating more SAR target features at different azimuth angles is beneficial for the prototype calibration. When the number of generated support features is beyond 72, the classification performance of the model under cases 1 and 2 does not improve significantly and even declines in some cases, despite the increasing number of generated support features. Specifically, in the results of case 1 displayed in Fig. 13, it is obvious that generating 360 support features for each real support feature results in lower classification accuracy compared to generating 120 support features in the 5-shot case. This is due to the bias of the prototype resulting from the random azimuth angles. Reviewing the experimental results for case 2, we can see that the classification performance remains almost consistent when generating 72 support features or generating 120 and 360 support features. This is because the scattering centers of SAR targets remain stable over small azimuth intervals [80], [81], so generating support features with too small azimuth intervals does not increase the diversity of target information. Moreover, generating 120 and 360 support features for each real support feature significantly increases training time.

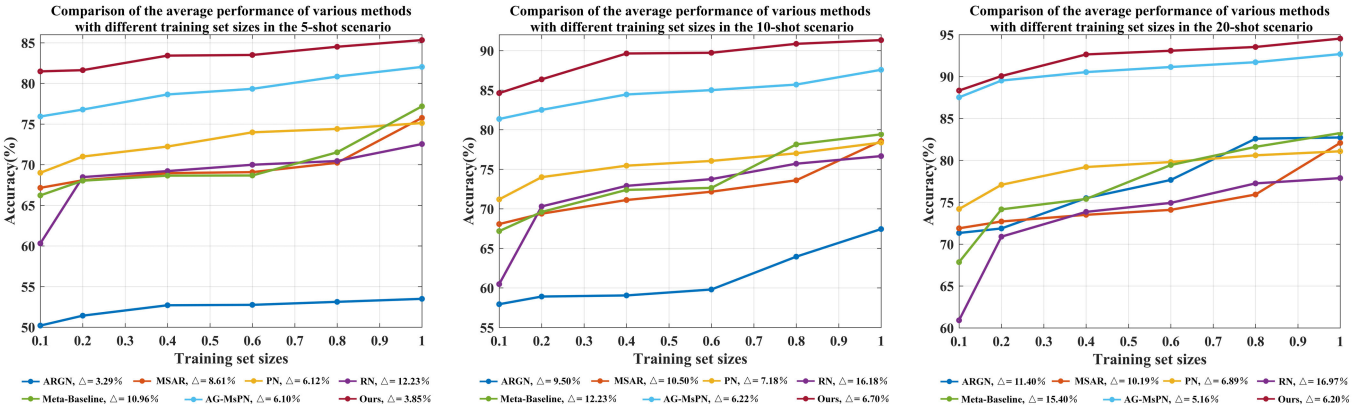
Furthermore, comparing the experimental results of cases 1 and 2, we can observe that the results of case 2 are slightly higher than those of case 1 when the number of generated support features is the same. This is because the azimuth angles of the support features generated under case 2 are uniformly distributed over [0°, 360°), and the prototype is calibrated by considering the target information under the full range of azimuth angles and is thus unbiased. As for case 1, the calibrated prototype may be biased to a specific range of azimuth angles due to randomness, resulting in a lower classification performance than in case 2.

After comprehensively considering the experimental results of cases 1 and 2, we finally choose to generate support features whose azimuth angles are uniformly distributed in [0°, 360°) with an interval of 5° for each real support feature to improve both the classification performance and the computational efficiency.

G. Effect of Source Domain Sample Number on LcFGC

All the above experiments are implemented with all training samples in the source domain. We conducted relevant experiments to verify the effect of the sample numbers of the source domain on the performance of LcFGC and other comparison methods. Specifically, we randomly select 80%, 60%, 40%, 20%, and 10% from all samples in each class of the training set shown in Table II to form a new training set. Fig. 14 shows the average performance of LcFGC compared to other methods under various training set sizes, where each model was trained 20 times, and the support set of each experiment is different. Since the training of the comparison methods A-ConvNet and DA-Net are performed without samples of the

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5-shot 10-shot 20-shot

Fig. 14. Average performance of our method compared to other methods under various training set sizes, where the Δ in the legend represents the decrease in classification accuracy when the size of the training set is reduced from 1 to 0.1.

TABLE VII
TARGET DOMAINS FOR RESELECTION

Target Class	T1					T2			T3		
	BRDM2	T62	T72			2S1	D7	T62	2S1	BRDM2	ZSU23/4
	E-71	A51	SN132	SN812	SNS7	b01	92v13015	A51	b01	E-71	d08
Support set	298	299	232	0	0	299	299	299	299	298	299
Test set	274	273	196	195	191	274	274	273	274	274	274

source domain, the classification results of these two methods are not displayed in Fig. 14.

As shown in Fig. 14, the classification performance of our method and other comparison methods have different degrees of degradation when the number of training samples gradually decreases. For example, when the proportion of randomly selected training samples decreases from 100% to 10%, the classification accuracy of our method decreases by 3.85%, 6.70%, and 6.20% in the 5-, 10-, and 20-shot cases, respectively. The classification accuracy of PN drops by 6.12%, 7.18%, and 6.89% in the 5-, 10-, and 20-shot cases, respectively. For MSAR, there is a decrease in classification accuracy by 8.61%, 10.50%, and 10.19%, respectively. The variations of the classification performance of the other comparison methods according to the training set sizes are shown in Fig. 14, and more details of their results can be found in Fig. 14. Clearly, our method's classification performance still maintains the highest under different few-shot classification settings. Moreover, in the 5-shot case, the robustness of our method is only slightly inferior to ARGN, while in the 10- and 20-shot scenarios, the robustness of our method is only slightly worse than that of AG-MsPN.

Why is the classification performance of the proposed LcFGC incorporating FGAN not significantly deteriorated by the reduction of training samples in the source domain? This is because we use a "task-driven" episodic training strategy. In the episodic training process, the classes and samples in the source domain are randomly sampled to form a variety of subtasks that mimic the few-shot feature generation and classification in the target domain. Even if the samples in the source domain are heavily reduced, LcFGC is richly experienced by continuously learning different subtasks, thus generating high-quality support features for unseen classes to calibrate the prototypes. Specifically, in the three-class target classification task, the training set of the source domain (containing seven classes of targets) and the support set of the target domain (including three classes of targets) form a complete training set with ten classes of targets to train our model. Then, we can acquire $C_{10}^{3} = 120$ different subtasks without repeated sampling. Therefore, even though the number of training samples in the source domain is reduced, the classification performance of our method does not drop dramatically owing to the numerous and diverse subtasks learned during the training process.

H. Generalization of LcFGC

The experimental results in Table IV show that the proposed LcFGC achieves state-of-the-art performance under the division of source and target domains in Tables II and III. In this section, we divide the MSTAR dataset into multiple target domains according to the settings in [70] to thoroughly validate the generalization ability of LcFGC. Table VII reports the support and test sets for the T1, T2, and T3 target domains. Then, for each target domain, the samples with a 17° depression angle of the remaining classes compose the training set of the corresponding source domain. The average classification results of LcFGC and other comparison methods under different source and target domains are displayed in Table VIII, where each model was trained and tested 20 times.

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TABLE VIII
CLASSIFICATION PERFORMANCE OF COMPARISON METHODS UNDER DIFFERENT TARGET DOMAINS

Method	T1			T2			T3		
	5-shot	10-shot	20-shot	5-shot	10-shot	20-shot	5-shot	10-shot	20-shot
A-ConvNet [1]	61.80%	69.34%	76.78%	58.36%	70.00%	78.33%	72.79%	78.25%	84.04%
DA-Net [20]	62.96%	68.85%	76.57%	59.88%	68.99%	76.20%	73.39%	77.32%	83.74%
ARGN [24]	60.11%	71.72%	85.47%	61.09%	81.98%	92.09%	68.53%	84.50%	93.31%
MSAR [34]	88.30%	90.88%	92.08%	89.74%	92.11%	93.52%	91.60%	93.19%	93.81%
PN [36]	85.71%	88.03%	90.44%	89.51%	91.78%	93.71%	89.57%	90.83%	91.12%
RN [37]	83.94%	85.28%	86.28%	89.39%	91.08%	92.51%	89.01%	89.83%	89.92%
Meta-Baseline [84]	85.78%	88.77%	89.94%	88.92%	92.70%	94.16%	89.66%	91.08%	92.63%
AG-MsPN [70]	88.43%	91.09%	95.38%	94.60%	96.42%	98.06%	95.00%	98.05%	98.63%
Ours	**89.92**%	**92.24**%	**95.71**%	**94.87**%	**96.67**%	**98.23**%	**95.27**%	**98.08**%	**98.82**%

TABLE IX
RUNNING TIME OF DIFFERENT MODELS

Method	5-shot	10-shot	20-shot
A-ConvNet [1]	0.0023s	0.0023s	0.0023s
DA-Net [20]	0.0019s	0.0019s	0.0019s
ARGN [24]	0.0012s	0.0013s	0.0015s
MSAR [34]	0.0024s	0.0024s	0.0024s
PN [36]	0.0037s	0.0037s	0.0046s
RN [37]	0.0038s	0.0041s	0.0047s
Meta-Baseline [84]	0.0182s	0.0222s	0.0288s
AG-MsPN [70]	0.0085s	0.0096s	0.0127s
Ours	0.0120s	0.0132s	0.0226s

As seen from Table VIII, the performance of the proposed LcFGC with different divisions of the target domain is significantly superior to the performance of comparison methods except for AG-MsPN in the 5-, 10-, and 20-shot cases. For AG-MsPN, which also has high classification performance, the classification accuracies of our method are 1.49%, 1.15%, and 0.33% higher than AG-MsPN in the 5-, 10-, and 20-shot scenarios, respectively, under the target domain T1. For the target domains T2 and T3 without variants in the test data, since the prior attribute knowledge of SAR targets is applied in AG-MsPN, the classification performance of AG-MsPN far exceeds the performance of the other seven comparison methods. In comparison with our method, which does not utilize prior attribute knowledge, the classification performance of AG-MsPN is still inferior to ours. The above phenomenon confirms the excellent generalization of the proposed LcFGC under different divisions of source and target domains.

I. Running Time Analysis

In this section, to verify the real-time performance of the proposed LcFGC, the running time during the test procedure has been carefully analyzed. Concretely, to fairly show the running time of the comparison methods during the test procedure, we feed the samples in the test set shown in Table III into the comparison models with the same batch size (batch size is set to 1). Each model is tested 20 times, and the average running time of each sample in the test procedure is shown in Table IX. All the experiments are implemented on a DELL workstation with the Intel Xeon Silver 4210R CPU (2.40 GHz) and the NVIDIA GeForce RTX 2080 Ti GPU (11 GB memory).

As shown in Table IX, the test time of the lightweight models proposed in the early years (i.e., A-ConvNet and DA-Net) is short, and the running time of ARGN is also fast due to the support vector machine [86] used in the test procedure. For the metric-based meta-learning models PN, RN, and AG-MsPN, the correlations between the sample features and the prototypes of each class need to be calculated, and the test time is therefore increased. The optimization-based meta-learning model MSAR uses fully connected layers to directly predict the classes of the samples, so the test time for MSAR is slightly lower than those of PN, RN, and AG-MsPN. In the proposed LcFGC, FGAN and AFCM are required to generate and correct the support features in the test procedure. Hence, the running time of LcFGC increases but is still lower than that of Meta-Baseline. Moreover, we think that the increased running time is acceptable in exchange for the significantly improved classification performance.

V. CONCLUSION

In this article, we propose LcFGC, which consists of MLC-Net and FGC-Net, to enhance the performance of few-shot SAR target classification. First, to highlight the contribution of vital local features of targets on SAR images to classification, we design MLC-Net, a high-performance few-shot classifier based on local feature classification. Considering the degradation of classification performance due to the lack of SAR target images in lots of azimuth angles, we propose FGC-Net to generate and calibrate the support features of SAR targets in the full range of azimuth angles to complement the real support features extracted by MLC-Net. Experiments on the MSTAR dataset validate the effectiveness of the proposed LcFGC. In the future, we will focus on incorporating domain knowledge of SAR images to aid in the few-shot SAR target classification.

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